

David Feldman

Bay Area, CA | dodya.feldman@gmail.com | github.com/dodyya | linkedin.com/in/dodyya/ | dodyya.github.io/

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

B.S. Mathematics and Computer Science - GPA: 4.0

Expected Grad May 2029

- State Farm CS Scholarship: 1 of 2 from a cohort of 2,500+ freshmen recognized for exceptional academic promise
- Coursework: Computer Architecture, Algorithms, Data Structures, Linear Algebra, Probability, Computational Geometry
- Societies: James Scholar Honors Program, Quant @ Illinois, SIAM, ACM SIGma, SIGPLAN

RESEARCH

Illinois Mathematics Lab

Jan 2026 — May 2026

UIUC Math Department

Urbana, IL

- High-performance Rust and Python tools to compute the EHP Spectral Sequence, an object of study in algebraic topology
- Computed farther than original paper, created the most comprehensive visualizations to date, presented poster ([repo](#))

SELECTED PROJECTS

Real-Time Audio Visualization Suite

- Rust implementation of iterative Cooley-Tukey FFT from scratch; real-time spectrogram with synchronized WAV playback
- Audio/video sync; memory-mapped parser with zero-copy reinterpretation and bidirectional sliding-window cache

High Performance Particle Physics Engine

- GPU-accelerated Rust physics engine simulating kinematics and particle interactions for 120,000+ particles at 60 FPS
- CUDA kernels with cuda-oxide, device-side OpenGL ↔ CUDA interop, spatial partitioning for cheap particle interactions

Interactive Piano-Mover's Problem

- C++ computational geometry motion planner for a translating and rotating 2D polygonal robot: 3D path in free space
- Construction of (x, y, θ) configuration space by Minkowski sum of robot with obstacles; tetrahedral subdivision, Dijkstra

Eulerian Fluid Simulation

- Interactive real-time 2D CFD simulator with incompressible flow, obstacle interaction, and multiple visualization modes
- MAC grid velocity field with backward advection sampling, divergence-free projection for pressure solving in Rust

Automatic Differentiation Engine

- Reverse-mode automatic differentiation engine in Rust; topological DAG sort for the gradient of an arbitrary expression
- Demonstrated with end-to-end neural network training via backpropagation

Constraint Satisfaction Puzzle Solver

- [Flow Free](#) game clone and solver in Rust, backtracking and forced move greedy heuristics, Python puzzle collection pipeline

End-to-end Mechanical Keyboard

- Fully custom printed PCB with QMK/C, community group buy and open-sourced case hardware and firmware

SKILLS

- **Languages:** Rust, C/C++, Python, MIPS Assembly, OCaml
- **Tools:** Git, Cargo, CMake, cargo-flamegraph, samplify, rayon, PyTorch, NumPy, KiCAD
- **Concepts:** Numerical Methods, Optimization, Computational Geometry, Concurrency

EXPERIENCE

Full-Stack Developer

Sep 2023 — May 2025

Los Altos Hacks

Los Altos, CA

- Developed software infrastructure for the world's largest high school hackathon for two years in a row
- Coordinated feature development on multi-team React/TypeScript projects through Git workflows
- Led the judging system sub-team, implementing custom pairwise voting to streamline in-hackathon voting